

MEC4407 / ENG5105: Mechanical Discipline

Project Brief



The world's most powerful consumer-grade 3D printer hotend

Project Developed by:

Scott Wordley

Kathy Petkoff

Live Version from: Jul 25, 2023

IMPORTANT NOTES:

Please do NOT download this document and save it offline.

We will update this shared online version with comments, clarifications and new information throughout the project. Please always open, review and refer to this online version. It is a living document.

Special Consideration / Alternate Assessments:

Please note that due to tight and fixed timelines for our oral presentations and also the review of your projects and feedback, **we will not be able to provide any extensions for the team submissions, or for the oral presentations of team members not individually impacted by events which qualify for special considerations.**

We will only be able to reschedule (at a later date) the oral presentations (only) for individual students who are impacted by events which qualify for special considerations. We would rather not do this if it can be avoided, as you miss the opportunity to present to your peers. They will be given the team marks for any meetings they are unable to attend, and extensions on their Lab submissions.

What this means is that if a team member is suddenly unable to contribute to the project, the team will need to redistribute and cover their responsibilities. This is exactly what would happen in the real world. We have provided teams with a large number of members to allow the load to be more easily shared. It is the responsibility of teams to manage the early drafting and delivery of their projects such that they are not caught short by missing contributions close to deadlines.

1.0 The Project

The major project will challenge teams of 4-6 (nominally 5) students to ***design, analyse, manufacture and test the world's most powerful consumer-grade 3D printer hotend.***

We have chosen to approach this project using role play, to simulate a typical client / consulting engineer dynamic that is common in industry. This is likely very different to your experience of university projects so far, where you are usually taught and given everything you need, and then allowed to work. For this project teams will need to interact with staff to uncover the client's requirements, determine what is possible based on your skills and resources, and then negotiate and agree to the engineering services that will be delivered.

We have chosen a complex, open-ended, multi-disciplinary and authentic design project. Teams will be encouraged and supported to renegotiate and expand their project scope by identifying other opportunities for design and implementation within their allocated spheres of project influence. The mentors may have particular areas of interest or expertise that they would like teams to explore, and teams should be responsive to such advice or interest. Teams and members will need to think and act like a company of consulting engineers who are tasked with delivering a real world project outcome, and not like a group of students who need to be told exactly what to do at every turn. Work hard to impress your client with your teamwork, skills, and professionalism and deliver value for the project.

2.0 The Scenario

You are a recently graduated Mechanical Engineer, who has just secured a job at a top local Engineering consultancy firm. You have done your initial inductions with the company and are about to be allocated your first client, job and group of graduate teammates to work with.

Our MEC4407 unit coordinator Scott Wordley will play the role of **the client**. This client is a CEO of a large, local 3D printing company who has money to spend on consultants to undertake benchmarking and engineering development of improved hotend designs.

Our ENG5105 unit coordinator Kathy Petkoff will play the role of a **principal (or managing) engineer** at your company and the manager of your team of graduate engineers. Like every manager she is busy so she relies on the team to do most of the legwork.

Scott will make it clear when we would like everyone to act "in character", which will most commonly be for portions of the weekly Tuesday Workshops. Outside of these times Scott and Kathy are both available to help and support you like they normally would.

3.0 Our Teaching Team

Scott Wordley MEC4407 Unit Coordinator scott.wordley@monash.edu	Director of Student Teams, Lead Designer of the Faculty of Engineering Makerspaces and M+AE Student Workshops. Academic Advisor to Monash Motorsport, Monash Forge, Monash BrewLab, Precious PLastic Monash. Unit Coordinator of MEC4407 since 2009.
Kathy Petkoff ENG5105 Mechanical Discipline Coordinator kathy.petkoff@monash.edu	Mechanical Engineer from Industry with more than 15 years experience. Former Faculty Advisor for Monash Forge and Precious Plastics Monash, teaching and advising students for 7 years through their design and capstone project journeys.
Zachary Neilson Studio Manager zach.neilson@monash.edu	Completed my FYP with Scott on setting up 3D printers within the makerspaces. I have spent my life working on cars with my dad along with a year on the MMS team in 2017 working in the Chassis section. We won Auscomp with both our electric and combustion vehicles that year. Experienced in manual machining and fabrication techniques. Experienced in 3D Printing. Currently managing the Design and Build Studios.
Bryan Liew Studio Manager Bryan.Liew@monash.edu	Bryan was a member of Precious Plastic Monash as Technical Member 2021-22. Extensive knowledge of Solidworks CAD, 3D modelling, 3D printing. Experience with manual machining, and fabrication. ENG1011 and MEC2404 demonstrating experience.
Lachlan Cunningham lachlan.Cunningham@monash.edu Demonstrator	MUAS Aerostructures section leader from 2019-2021. Design experience predominantly involving composite structures and aerodynamics of small (<10kg) aircraft. Currently completing a computational fluid dynamics PhD, simulating intense winds and complex flows generated by bushfires (such as fire-tornados)
Jeremy Pryor Jeremy.Pryor@monash.edu Demonstrator	MHPR Aerostructures section lead 2020-2021. 2nd Year PhD Student modelling Rotating Detonation Engines (RDEs) both experimentally and numerically. Student machinist with both manual and NC machining experience in 3 and 4 axis.
Dhiluka Mapa dhiluka.mapa@monash.edu Demonstrator	MNR chassis lead from 2022-23 and CTO from 2023-24, 4th year Mechatronics student. Designed and manufactured the Rover's current drive system
Redmond Henry Redmond.Henry@monash.edu Demonstrator	4th year Aerospace Engineering Student, Monash HPR Operations lead 2024-2025. COO of Monash HPR, Designed and manufactured components for High Powered Rockets

4.0 The *Concept Design* and Reporting Process

Teams will work towards two major design gateways, and receive feedback and recommendations from the Client at the conclusion of the first of these gateways. The first gateway is the **Concept Design Submission**, which is detailed in this document. The second major gateway is the **Detail Design Submission**, which we will provide additional details for later in the semester.

Please keep referring back to this live document, we may make minor updates to the report requirements.

Important Information:

- Teams will be asked to benchmark a competitor's hotend and report these results, before undertaking their improved design for the client (specifically the new **Prusa i3 MK4 hotend**). This is very common in industry, to validate your test and analysis processes and to establish a level of performance to compare against.
- Teams should assume that you are consulting to a company, that wants to improve the hotend design for their *standard* [Slice Mosquito Hotend products](#) that they use. (Note: Not the Magnum, Magnum+, Liquid or Conduct variants).
- You have been asked to *retain the current heatbreak architecture in this Mosquito product, while designing an improved hotend component, which is able to accept one of the three current standard nozzle sizes* ([Standard length, Volcano or Super Volcano](#)).
- Teams will utilise a standard BondTech extruder, connecting to the Slice Mosquito heatbreak (**exact model to be specified, likely the DDX**), to control the filament drive.
- Your revised hotend should take advantage of a *single* (newly released) [75W heating cartridge](#), in the standard size factor. This is a significant power upgrade over the standard 40W cartridge.

[Please aim for 35 pages MAX for sections below + References + Appendices]

In the concept design stage, teams are required to document (under the suggested headings / template / rubric):

[1 page] - Title Page with Team Member Names and Headshots

[1 page] - Executive Summary

[1 page] - Acknowledgement of Country

[1 page] - Table of Contents

[~4-6 pages] - Introduction

- Teams should clearly summarise and reflect back their understanding of the needs and wants of the client for this project;
- Teams reference and summarise the key published research and grey literature on the project topic.
- The team should work to distil out an initial table of requirements and specifications for the project (Refer to [MEC2402](#)) based on the customer's emails and ongoing discussions in class.

[~2 pages] - Project Scope and Timeline

- Clearly **define the tasks required (for the first phase of the project) in a chronological order** and explain the team's approach to the project scope.
- Teams should confirm their specific allocated hotened benchmarking and design scope with regards to:
 - The hotend the client would like experimentally and analytically benchmarked for performance.
 - The heating element type and number of heating elements that the client would like the team to design a new hotend for.
 - The heatbreak and extruder that the client would like the team to design for.

[1 page] - UN Sustainable Development Goals

- Review the [UN SDGs](#) and **identify one goal that the team considers is the most relevant to this project**. Teams should connect the outcomes of the successful delivery of this project with the targets and indicators for this SDG. How does the delivery of this project deliver impact or help us progress towards this goal?
- This is a good place to deploy GenAI to support your work (but you may use it elsewhere).
- Please make use of GenAI tools and appropriately acknowledge them based on the LearnHQ advice and recommendations in our workshop slides.

[~6-8 pages] - Benchmarking: CAE Analyses

Based on your ongoing consultation with the client, present CAE simulation results for a relevant market competitor (**all teams will benchmark the new Prusa i3 MK4 hotend**).

These analyses should include:

- Hand calcs and/or Structural FEA to assess mechanical strength of the hotend;
- Hand Calcs and/or Thermal analysis (using the ANSYS Mechanical / Structural approach) to assess performance of the heat break and maximum flowrate of the hotend.
- **CFD, and CFD Thermal analysis is not required for this concept stage (but will be for the Detail design phase).**

For the thermal simulations, teams should try to predict analytically: the heat break performance (consider a temperature versus Z distance graph to plot this gradient) at constant 230 deg thermistor surface interface temp; and the maximum PLA volumetric flow rate possible (given a maximum possible heating rate of 40W), that maintains a 230 deg temp at the thermistor interface surface. It may be possible to plot a graph of PLA volumetric flowrate versus average heater power input.

The methods used for each approach should be detailed and summary results presented (full simulation reports can be included in Appendices). Teams must present empirical hand calculations (using spreadsheets) for their structural and thermal modelling/analysis of the hotend and heat break performance. These hand calc predictions should be compared and contrasted with the finite element thermal computational results (and the experimental benchmarks made below), with any differences discussed and explained.

[~2-6 pages] - Benchmarking: Physical Testing

Based on your consultation with the client, present physical test results for your benchmark hotend (**the Prusa i3 MK4 hotend**). These tests should interrogate the performance of this hotend, and you must state a maximum “practical” printing flowrate (with reasonable error limits $\pm$ stated based on your experimental uncertainties), taking inspiration from and making required adjustments to any processes that you can find in the grey literature (an upcoming Workshop activity). Experiments should be designed and conducted in the prac times using the printers we have available, and the method and summary results only presented in the body of the report (full experimental reports can be included in Appendices). Thermal imaging or probing may be possible for validation of these tests but is not essential, please explore with the client and studio staff.

[~2-4 pages] - Creative Solution Generation

Please describe the team’s approach to generating a large number of creative solutions to your design problem (within your given design bounds as set by the client) using divergent thinking methods or techniques. A selection of outcomes from this process can be included in the report, and the full range in appendices.

[~1-3 pages] - Concept 1: Concept name

- Present the details for the first of two **high level concepts** (inc. a CAD Assembly and annotated, exploded Assy). Why is this concept one of your preferred options?

[~1-3 pages] - Concept 2: Concept name

- Present the details for the second of two **high level concepts** (inc. a CAD Assembly and annotated, exploded Assy). Why is this concept one of your preferred options?

[~2-3 pages] - Decision Making

- Decide and explain which concept is the most preferred, i.e. **decision making** (use of formal technique like composite criterion, is encouraged, refer to [MEC2402 content](#)). Rank/Score these concepts against your (and the client's) requirements and specifications.

[~1 pages] - Plan for Further Work

[No page limit] - Acknowledgements

It is a requirement that each team make use of GenAI tools to improve their reports. You will need to correctly acknowledge this use, by following the recommendations and examples hosted at Monash's LearnHQ site, and our recommendations in the workshops. Please see staff for feedback in advance of the submission if you have any questions or concerns.

[No page limit] - References

[No page limit] - Appendices

- ~~Appendix A – Contract of Work~~
 - ~~Please provide your Contract of Work (as developed via the W2 Studio activity)~~
- **Appendix B etc as required for test reports etc**

Teams will submit a written **Concept Design Submission** in W5 and deliver their **Concept Presentations** on the same topic in W6 in the Prac class times.

Further details will be released for the requirements of this Concept Design Oral Presentation, and the second Detailed Design Oral Presentation.

Collectively, the clients and your fellow “pod” teams will review these concepts and together you will synthesise a single, final **Concept Design**. Together you will work to refine this design, analyse it, and determine how it will be manufactured and then tested as part of which is the **Detail Design Phase**. Additional details are provided below for the requirements of that second stage.

Teams will submit a **Detailed Design Submission** in W11 and deliver their **Detail Design Presentations** in W12.

5.0 The *Detail Design* and Reporting Process

The second and final gateway is the ***Detail Design Submission***, which is detailed below. This *work* should be undertaken in partnership with the other teams in your designated pod of 4-5 teams. Each pod will be provided with a new shared drive that they can use to collect and share the work of the pod. Each team will need to *write* and submit their own final report, as per the requirements below. Teams in the pod may share the same test and analysis reports (worked on collectively), which can be included in the appendices.

This report may contain any of the shared work completed by your fellow pod members (CAD, analysis, testing etc) but the writing in the main body of the report must be done by each team in isolation, with these reports located on the team's individual drive folder, and not shared with, or copied by the other teams.

Each team will provide an oral presentation which touches on and summarises the entire scope of work for this phase of the project.

We expect the teams in each pod to work out a way to distribute the work fairly amongst teams and members, and review your progress regularly (at least weekly) to ensure that all requirements are met and that all teams have an appropriate understanding of the work that they will present. We expect this to challenge your teamwork and management skills, but the better your organisation and management, the more efficient this process will be.

[Expected Max of ~45 pages + References + Appendices]

Report Overview: **[Total 50%]**

[~4-5 pages] Title page, Executive Summary, Acknowledgement of Country, Table of Contents **[1%]**

[~2-3 pages] Summary and consolidation of Pod Hotend Design Concepts **[1%]**

[~2-3 pages] Final Hotend CAD Design **[1%]**

[~1 page] Hotend Material Selection **[1%]**

[~4-6 pages] Hotend Thermal Analysis **[6%]**

[~4-6 pages] Hotend Structural Analysis **[6%]**

[~4-6 pages] Part Cooling Fan Duct: Design and Fluid Analysis **[6%] NOT Required for Masters Pod**

[~1 pages in report] Engineering Drawings **[4%]**

[~4 pages] Manufacturing Proposal (CAM) for the Hotend **[6%]**

[~2 pages] Manufactured Hotend Photos **[1%]**

[~2 pages] Costing and Recommended Retail Price **[1%]**

[~2-3 pages] Part Cooling Fan Duct(s) Manufacturing **[4%] NOT Required for Masters Pod**

[~4-6 pages] Maximum Practical Flow Rate Testing **[8%]**

[~1 pages] Conclusions **[1%]**

Report Structure, Formatting, Grammar and Expression, Figures, Tables, Refs **[3%]**

Detailed Advice:

Teams are required to document under the suggested headings / template / rubric:

[1 page] - Title Page, Team Member names and professional headshots, pronouns

[1 page] - Executive Summary

[1 page] - Acknowledgement of Country (please refine this from the Concept Report)

[1-2 pages] - Table of Contents [1% for this and above content]

[~2-3 pages] - Summary and consolidation of Pod Hotend Design Concepts [1%]

- Teams should consider the best concepts presented by teams in their pods and identify the preferred concept, and/or features from multiple concepts.
- A formal decision making is not required, instead we suggest that teams work to synthesise a single design, which adopts the best features from the concepts presented.
- Please note that this understanding may evolve and change over the course of the project based on your work and analysis and testing results and also your in person discussions and consultations with the client.

[~2-3 pages] Final Hotend CAD Design [1%]

- Use CAD images to present the detailed design of your Hotend assembly, including the Slice Mosquito heatbreak, the specified heater cartridge (with part of the cable shown), the specified thermistor (with part of its cable shown), and any required fasteners (bolts, nuts, grub screws etc)
- Please clearly show:
 - this entire assembly, including your hotend and nozzle, the Slice components utilised and all fasteners (multiple views). An exploded view may be helpful.
 - the hotend component in isolation (multiple views).
- Items that should be excluded from this design are:
 - The Bondtech DDX extruder;
 - The heat break fan;
 - The part cooling fans and ducts;
- Please provide a hyperlink to your CAD assembly of all these components, saved to the pod's Drive folder.
- Please briefly discuss your design process and key features of the hotend.

[~1 pages] Hotend Material Selection [1%]

- Present a table with your top ~5 or 6 hotend material options and the most relevant material properties for this component. Based on what you are trying to achieve with your design (refer to client concept feedback), make a case for your most preferred material choice. It is ok if this choice is different to the material selected by teaching staff for the manufacturing, due to time and availability constraints. If relevant you may refer to later analysis to justify your selection (i.e. thermal or structural).

[~4-6 pages] Hotend Thermal Analysis [6%]

- Conduct thermal analysis of your hotend design using ANSYS (structural approach recommended). Hand calcs are not required for this submission.
- You should report on the heatbreak performance of your new design.
- Clearly explain your thermal analysis setup, particularly your convection surfaces and coefficient values.

- Assuming a thermistor temperature of 230 deg C (closest to the nozzle), estimate the maximum PLA flowrate of your hotend. Please have your pod confirm this prediction (by emailing the Scott and Kathy) before you commence physical testing of your hotend, so we can gauge the resulting accuracy of your predictions.
- At max flowrate, estimate the total amount of power input required from your heating cartridge, and the proportions of power 1) used to melt the filament and 2) lost to conduction/convection.

[~4-6 pages] Hotend Structural Analysis [6%]

- Conduct a structural analysis of your hotend using ANSYS. Hand calcs are not required for this submission.
- Clearly explain your FEA setup, particularly your supports and applied forces.
- Predict the most likely failure modes for the hotend component (itself) and the loadcases they are likely to occur at (direction and magnitude of forces).
- Contrast the structural performance of your new hotend with the hotend benchmarked initially. In the context of the full assembly (up to the attachment to the extruder), does your design experience failure at lower or higher forces? Quantify the relative magnitude of this difference as a % increase or decrease in your failure forces.

[~4-6 pages] Part Cooling Fan Duct: Design and Fluid Analysis [6%]

(This task is NOT Required for the Pod of Masters Teams, due to the smaller student numbers in this pod. These marks will be redistributed equally between the other major 6% tasks)

- Design a part cooling fan duct that:
 - For attachment and mounting, sandwiches between the BondTech DDX extruder and the heatbreak fan, using bolts which go into the three M3 threaded lugs.
 - Mounts two of the large fans (N052AR) for part cooling.
 - Allow for neat cable securing and management.
 - Provides appropriate ducting from the outlet of each of these fans to the nozzle of your hotend, to cool the extruded filament. This ducting must be located above the lowest point of the nozzle, lest it catch on the prints.
 - Can be manufactured using 3D printers available in the studios.
- Conduct a fluid flow analysis on your fan shroud(s) using ANSYS
- Explain your choice of overall fluid domain, wall boundary treatments, and how you have chosen to simulate the fan (inlet) and your domain outlet.
- Explain your meshing strategy, mesh metrics, and choice of turbulence model.
- Present the following cut planes (all perpendicular to the flow) and show contours of velocity, and separately, pressure contours.
 - Just beyond the fan inlet;
 - At about four locations evenly spaced along each fan duct;
 - At additional locations relevant for filament cooling
- Comment on the flow performance of the duct(s), any issues identified (separation, uneven flow, stagnation regions etc).

[~1 pages in report] Engineering Drawings [4%]

- Complete a formal A3 sized detail drawing of your Hotend (heat block) component to AS1100.
- Complete a formal A3 sized assembly drawing of your hotend, including all of the Slice Mosquito components, your heatbreak fan, your part cooling duct(s) and part cooling fans and associated fasteners, but NOT the BondTech DDX extruder. A BOM table should be included.
- Please link to high quality, high resolution PDFs saved in your drive, from your report rather than insert them as lower-res images.

[~4 pages] Manufacturing Proposal (CAM) for the Hotend [6%]

- Use Fusion360 to CAM the manufacturing of your hotend component, assuming it will be manufactured using:
 - An OKUMA GENOS m460V 5-Axis Vertical machining centre (as in Studios LG08)
 - Lang Vice and Jaws (details to be provided)
- The following tooling can be assumed as available:
 - 50mm face mill (insert cutter)
 - 25mm face mill (insert cutter)
 - 12mm,10mm,8mm,6mm,4mm and 3mm End Mills
 - HSS drills from 1.0 to 13.0 in 0.1mm increments
 - 6mm spot drill
 - Standard metric taps
 - If you require ball nose cutters please consult with staff on available sizes
- Clearly specify your assumed starting stock size, shape, dimensions and material.
- Present the key manufacturing steps and settings and describe the machining strategies used, showing screenshots from Fusion360.
- Present a table showing all set-ups and major machining operations and the estimated time for each, and then total the time.
- You do not need to consider any coatings (that might be used in actual production).

[~2 pages] Manufactured Hotend Photos [1%]

- Present photos and a description of the process actually used to manufacture your hotend. Progress photos would be amazing if your team is able to coordinate with the machinist or have them take some pics themselves.
- These processes may differ from your CAM, depending on the actual CNC used by Bryce Noble (M+AE) who will be manufacturing your hotends.

[~2 pages] Costing and Recommended Retail Price [1%]

- Accurately estimate the cost of manufacturing your hotend, for a production run of 100 units, including:
 - Raw material cost (local market price)
 - Machining costs (assume AUD\$150 hour for CNC operator and factory overheads)
- Do not consider:
 - Coatings and associated costs.
 - Packaging and shipping costs.
 - Advertising, promo, web or social media costs.

- Based on an analysis of global market competitors propose a recommended retail price for your hotend.
- Graph your “break even” point for sales of this hotend and predict total profit if all 100 units were to be sold direct from the manufacturer (with no reseller margins).

[~2-3 pages] Part Cooling Fan Duct(s) Manufacturing [4%]

(This task is NOT Required for Pod of Masters Teams M1-M3, due to the smaller student numbers in this pod which has only three teams. These marks will be redistributed equally between the other major 6% tasks)

- Present photos and a description of the process used to manufacture your Part Cooling Fan Duct(s) using 3D printing (tech to be determined by team).
- Explain your slice setting, print orientation, support materialism infill% etc.
- Explain any post processing required.
- Show photos of the final part in isolation and assembled onto your printer head.

[~4-6 pages] Maximum Practical Flow Rate Testing [8%]

- Work with teaching staff and other teams to develop a consistent test model and approach to estimate the maximum practical flowrate of your hotend, using the Creality CR-10 S5 printers.
- Check and confirm that your experimental approach is not being limited by slicer or printer limits.
- Present images which show the onset of print defects which occur beyond the maximum print flow rate.
- Compare (using a bar graph) where your max flow rate sits relative to published maximum flow rate performance specs of your market competitors (Slice, Takoto, Dyze Design).
- Compare (using a scatter plot) where your hotend sits relative to your market competitors (Slice, Takoto, Dyze Design) based on recommended retail price AUD\$ (on the horizontal axis) and published maximum flow rate performance specs in (mm³/sec on the vertical axis).

[~1 pages] Conclusions [1%]

- Consider the project as a whole when you draw your conclusions.
- What has the team learnt from participating in this project?
- What has the team achieved for the client?
- What are the (hypothetical) effects of this project?

Report Structure, Formatting, Grammar and Expression, Figures, Tables, Refs, Appendices [3%]

[No page limit] - References

[No page limit] - Appendices

~~Appendix A – Contract of Work [2%]~~

- ~~• Please provide a Contract of Work (to cover the work contained in the Detail Design phase)~~
- ~~• Please price your labour assuming that your team members have completed all the work on their own (without the help of your pod mates).~~
- ~~• Each team should create their own contract of work please, rather than share a single version created by the pod.~~

Teams will submit a written **Detailed Design Report** in W11 and deliver their **Oral Presentations** in W12.

6.0 Team Formation and Teamwork

Given the project's large scope and potential depth, the teaching staff will allocate approximately 4 - 6 students to each design team. Teams will be formed by the teaching staff from students within the same Practical class times. We will work to try and ensure that teams will contain an evenly distributed mix of on-campus and remote students, where possible.

All project discussion, communication and documentation is to be conducted in English. Please install the English language versions of our software to assist our teaching staff supporting you. International students: your English is typically fantastic, we want to assist you to develop your confidence in using it in professional workplace contexts.

The **ITP Metrics peer evaluation tool** will be utilised within each team to provide feedback on teamwork, and potentially to also scale the team marks for individuals.

7.0 Project Data Pack

Our design teams will meet with the client regularly and receive additional documentation and information relevant to the project, to assist all teams. Teams will also contribute their own research to this data pack. This data pack will most likely be updated throughout the course of the project as new questions for information are received and filled. Please check the Drive Folders regularly (which is where all this info is stored), and watch for our announcements and updates about new content. Do not download all this information in W0 and expect that you have the complete set, always refer to the live google drive folders.

8.0 Moodle Access

Moodle access will be provided for all students (by request if not available) to access foundational design knowledge from prior units. Students in ENG5105 in particular should refer to these units as the generic design process and machine element design skills that will be knowledge assumed by this unit and the project:

- MEC2402: Design Methods 2021 S1
- MEC3416: Machine Design 2020 S2

9.0 Client Requirements and Requests

We are going to roleplay a lot of elements of this project, to give you a more authentic experience of how the engineering industry actually works. Please come along for the ride and try to act as you would in the real work as a new graduate engineer, we are testing your professionalism and job-readiness. This extends to all aspects of your work, including your email communications, your chat messages with team mates, and the way that you interact in person with team mates and staff.

To set the scene, your design team receives the following (somewhat cryptic) email chain, which is typical of what happens in industry! Your first task will be to help the client expand upon their needs and wants for this project, before you can start work.

We will explore this together, actively, in our first Workshop in W1.

Client Questions and Answers:

We will periodically take and answer questions to the client in Workshop times, [please add your questions and review our responses here.](#)

From: scott@3dprinting.com.au

To: Kathy@monashconsulting.edu

Subject: 3D Printer Hot End Project Idea - Can we meet to discuss?

Hi Kathy,

My name is Scott, and I am the founder and CEO of Australia's largest 3D printing business (3dprinting.com.au), and I have some ideas for a project that I would like to discuss.

I can explain more next week, but basically:

- We are getting more and more requests for very large jobs, what we would classify as super-large format, up to 1m x 1m x 1m.
- For this size print, we have a fleet of mainly [Modix](#) machines now, and we plan to buy more soon. See attached picture.
- We find the standard [E3D hotends](#) that they come with to be unreliable, and are keen to find a better option.
- We have done some in-house testing on a range of different [Slice Engineering](#) hotends from the USA, and have been pretty impressed with their design and performance.
- We also came across some interesting units made by a small local company called [Takoto](#) which have some cool features and look like they also have potential. We have a couple of samples arriving soon.
- We really like the [BondTech](#) direct drive extruders, so are keen to try to figure out something that works with them.
- We would like to do some proper performance testing on some of these commercially available options that we have on-hand, and then have your engineers come up with some ideas on how we might improve these current designs even further?
- Our prints are so big that they take days and sometimes weeks to complete. If we can manage to get a 20% or maybe even 50% increase in flowrate of filament then we will be able to turn around jobs much quicker and get more work done.
- 3D printers of this size are a huge investment and they take up a lot of space in our facility, but the hotends are a relatively small cost (a couple of hundred \$). I would be happy to pay twice as much for double the performance, it would save me thousands each year!
- I know you engineers can produce lots of numbers and colourful pictures, so I will definitely need your guidance on the best way to approach this testing, design and analysis.
- If the \$\$\$ numbers all look good, I would like you to build a few prototypes and test them, to know for sure how they perform.

How does that all sound? Can we catch up next week to discuss via Zoom? Please bring your design team along!

Thanks!

Scott - CEO and Founder

3dprinting.com.au

From: Kathy@monashconsulting.com

To: scott@3dprinting.com.au

CC: grad_design_team@monashconsulting.com

RE: Subject: 3D Printer Hot End Project Idea - Can we meet to discuss?

Hi Scott,

Thanks for contacting us.

That sounds like a really interesting project, we would love to discuss it further.

Unfortunately, I am away all this week in Sydney for some professional development, so I won't be able to undertake my usual level of preparation for this meeting.

I am looping in my grad design team and will ask them to undertake some background research into the links you have provided.

Team, can you also please make sure you are up to speed on how FDM 3D printers work, and understand the names of the different components and terminology etc before our meeting? Why are some hotends better than others?

I am also certain I have seen some research papers online that undertake thermal analyses on 3D printer hotends, I just can't seem to find them right now... I trust the team will find them though!

If you can find some papers, it would be great to provide a summary or overview to Scott at our meeting, and we can all consider if we might be able to undertake some similar analytical approaches.

One other thing I did want to note, our company recently purchased some handheld [FLIR thermal cameras](#) which will be very helpful for correlation of any testing with simulation, so that is good news.

Thanks Scott and team, I look forward to catching up with you all on Tuesday next week!

Kind Regards,

Kathy

Principal Consultant

Monash Consulting

Week 2 Client Email:

Subject: W2 Meeting today! See you soon...

Hi Kathy and teams,

Thanks for agreeing to take on this project to develop a hotend for us, looking forward to catching up with you later today for a chat.

I would like to discuss and confirm a few things with you, primarily:

- I understand you have some **Prusa MK4 printers** there at Monash so we would like to have your teams **benchmark these new design hotends and heatbreaks** that they come with. Can you please **find out what the manufacturer claims for their performance** and then see if you can validate it with experimental tests. I saw a cool white paper from Slice Engineering on how they do this that you might use as inspiration? Beware the inbuilt firmware speed, accel and jerk limits on these printers, as they may cap the print speeds that you are requesting. There must be a way to check these are not impacting your tests, but I haven't figured it out yet sorry.
- Given you will have those experimental test results, it makes sense to me to have you develop an analytical approach using the same hotend design, to try and validate our numbers. So **please find and use the Prusa MK4 hotend/heatbreak geometry to conduct a structural and thermal analysis**. I am interested to know what kind of typical operation loads might result in structural failure. Thermally, it would be great to understand the heatbreak performance of the MK4 and see if you can get a flowrate prediction that matches your experiments. If we can do all that it will increase our confidence in the design work for the new hotend. If our new design can roundly beat these numbers we can be pretty sure we are onto a winner before we commit to a prototype.
- To fast track your new prototype design I would like you to **adopt the heatbreak used on the Slice Mosquito**, which is a really novel design. We have a number of them here and can provide for your use. Couple them with a BondTech DDX direct drive extruder. Then go about designing an improved hotend component below all that to work with the new 75W heater cartridges. Hopefully you can do better than the current Mosquito design and 50W cart, which is a little under powered for our needs.
- For these new **75W heater carts**, they have dimensions **Diameter 6mm and length 20mm**. I think let's have you work on designs that incorporate just one of these in the hotend. Please also assume the use of a single **PT1000 thermistor, diameter 3mm and length 15mm**. Perhaps stick to **off-the-shelf nozzle options (standrd, volcano or super volcano)**, as we tend to replace them regularly, so I don't want to source custom bits for these. Keen to see your concept ideas and justifications for the orientation and relative placement of these components in your hotends, get creative! Keen to see what your teams recommend.
- For prototype manufacturing please **work to the limitations of the CNC capabilities that you have there at Monash**. Any crazy ideas please run them past me for checking first! It would be nice to have the option of later doing a small production run (~100) for a reasonable price if we settle on a great design that works well. I will leave the choice of hotend material(s) up to you also.
- While ultimately we intend to run these on our fleet of Modix printers, to enable you to test and develop there at Monash I would recommend you **conduct your testing on your Creatily CR10 S5 printers** (when we get to the detail design phase). These are big enough to provide a good comparison. Check their power

supplies though, to see if they have enough to drive a 75W heating carts, or you might max them out?

- I discussed on the phone with Kathy that we would like to phase this project, so let's try to **work up a contract (and price) for this first phase of testing and concept design**. All going well we can run a second job for the detail design, first prototype and testing. Please provide an itemised quote, as I need to convince my board that this is a good investment!

How does that all sound?

See you on Zoom later today ;)

Thanks!

Scott - CEO and Founder

3dprinting.com.au

From: Kathy@monashconsulting.com

To: grad_design_team@monashconsulting.com

Subject: This week's Client Meeting - Thermal Approach

Hi Team,

Thanks for the chat last week regarding FEA and structural calcs. Hopefully you have a better idea of how to progress this analysis and benchmarking for the E3D hotend for your Concept Report. It is really important that we have a sound baseline to compare your redesigned hotend with.

In this week's meeting, I would like us to workshop **an approach to thermally model the MK4 hotend, using a spreadsheet and hand calcs, which can also be applied to ANSYS.**

Please consider the following questions as discussion starters:

- How do we model/calculate **conduction** and **convection**?
- What is the typical **ambient environment** that the hotend operates in?
- How can the **heat input** to the system be modelled?
- What **modes of heat transfer and heat loss** occur in our system?
- **How will you model these most critical modes?**
- What **other information** might we require about the hotend/environment to conduct these simulations/calculations?

I don't want full calculations or actual simulations, I just want to talk through your "modelling" strategy, before you make a start, to ensure you are all on the right track.

We will provide some images of the MK4 to help think this through.

Thanks team, we look forward to catching up with you all on Tuesday!

Kind Regards,

Kathy

Principal Consultant

From: scott@3dprinting.com.au

To: Kathy@monashconsulting.com

CC: grad_design_team@monashconsulting.com

Subject: Feedback on your Concept Designs and next steps

Hi Design Teams,

Thanks for your fantastic concept presentations last week, we are excited by the great work that you have done so far and the many creative concepts that have been proposed.

A summary of my feedback is presented below:

Structural Analysis:

It was reassuring to see that all teams identified the weak point at the neck of the heatbreak in the MK4 design. We have bent a number of these hotends at this exact spot due to crashes (mainly X or Y) and also whilst trying to change nozzles, without proper support. This gives me confidence that your analysis and assumptions are correct. We are interested to understand if the robustness of your new design will result in an improvement with respect to the MK4 benchmark, building off what should be a much more rigid and robust Slice Mosquito heatbreak assembly. ***Please present the structural performance of your new hotend, relative to the MK4 (how much better or worse % wise) for the main failure modes.***

Thermal Analysis:

We saw some great hand calcs and also ANSYS thermal modelling presented for the MK4. Given the Slice Mosquito has a more complicated thermal conduction path (heatbreak and four stand-off tubes) I am happy for teams to explore the heatbreak performance using ANSYS only this time. As discussed, please assume we are using the BondTECH DDX extruder, with its integral fan shroud and the small standard heatbreak fan. Again, please present your thermal performance relative to the MK4 benchmark. Based on the work we saw this week, the best measure of the thermal gradient would appear to be your predicted distance (in the heatbreak) between the glass transition temp (let's call it 60 deg C) and the fully melted temp for PLA (let's say 180 deg C). ***Please quote that distance for the MK4 (may require another look at your results) and the same distance for your new hotend/heatbreak and we should be able to infer any changes in print reliability (as a result of the heatbreak) from that.***

Hotend Material Selection:

Some teams presented material options and choices as part of their concept designs, but I wasn't convinced that the options had been examined rigorously and that teams were able to properly justify their selections. ***Let's meet next week to flesh out this***

choice with more hard data so we can make a good decision. Luckily this should not greatly affect the geometry development, material choice should be pretty independent of geometry (fingers crossed!). Keep in mind that we will need to source these materials as well, so try to find materials with local availability.

Final Hotend Geometry:

Working in your allocated pod of teams, please collaborate and try to converge on your preferred design. I am going to try and stay out of this process, to not spoil the fun! Given we have 7 pods all developing their own designs, I am content to see a wide variety of ideas put forward and developed as prototypes. Across that entire field, based on the results achieved we should be able to infer a lot about the physics of this problem and which design features drive high performance. We will not punish teams that select a lower performing concept, provided their engineering analysis and validation is of high quality. Understanding why something doesn't work well is incredibly valuable! *Please expedite this consolidation process, I would love to see where pods get to by Tuesday next week.*

Feedback on Hotend Concepts:

A couple of key things we noticed from team's concepts, to be aware of and ensure are correct for the next stage:

- Your hotend needs to accept the four stainless steel tube stand offs, that come standard on the Slice Mosquito. Double check their spacing and depth into your hotend please.
- To secure the hotend to the finned heat sink requires the four stand-offs (which react compression loads and bending and torque) but also two small screws. The standard Slice uses very tiny M1.4 screws, but there is a second option. The finned heat sink actually has a second threaded hole option spaced slightly wider that we will insist that all teams use. These threaded holes are M2.5. The studios also have a range of M2.5 screws in different (but very limited) lengths which can be used for this purpose. Please select an available size and design for that. In most cases this will require a certain thickness open tab with hole, or a counter bored hole made to the correct depth, to suit your screw, and supply the correct amount of thread engagement. Please carefully check the thread depth provided in the Slice heat sink as well, it is limited so these numbers need to be right, there is little margin for error.
- A couple of teams showed us concept designs with the Slice Mosquito MAGNUM model of hotend, which has a large melt zone and an external fibre insulator. Make sure this isn't your team! Some of the CAD for the Slice Mosquitos also looked a little off, with finns on the heatbreak tube being positioned too low, double check it. This is a common error from CAD found online.
- We strongly recommend that your selected nozzle its designed to tighten up and touch the lower face of the threaded heat break, so as not to leave any chamber or

passageways within the hotend. This allows for easy cleaning and clearing of stuck or jammed hotends. Adjust your sizings to achieve that please, double check the dimensions of your chosen nozzle accurately in person. We saw some very creative alternate nozzle designs from some teams but we would like to focus on the standard commercially available ones for this design please.

- We encourage teams to consider the manufacturability of their designs. We saw a large number of concepts with external radii and small internal corner radii, which could be a nightmare to machine. Jump into CAM and test your design and see if there are features that are problematic. Or ask for advice!

Part Cooling Fan Ducts (and mounting):

Given that your hotends are new and novel lengths we will need pods to also design and manufacture part cooling fan duct(s) and mounting (for the duct and the fans themselves). Have a look at a BondTech DDX and you will see that there are three threaded lugs available to mount the heatbreak fan. You are able to sandwich a plate in here (with a hole through the middle obviously for the fan flow) which will allow you to support your ducting and mount your fans (two of the big ones please). **Please design these ducts using CFD and then build them using 3D printing, they will be essential for cooling the filament flow out of your fancy new hotend, and ensuring the highest print quality and flowrate.** Just make sure they do not hang below your nozzle, or they will hit your prints!

Flowrate testing:

With your prototype hotend and part cooling fans and ducting in hand you will be able to test your creation. We have a Slice Mosquito for each pod to use, be careful not to lose any of these parts. You will be running them on our Creality CR-10 S5 printers, and these have upgraded power supplies which will allow us to crank more than 75W of heating, if required. **Please use 1.0mm nozzles for all your testing, for consistency.** Apply your lessons learned in the MK4 testing and let's see what you can do. We have the thermal camera as well to validate the thermals, so give that a shot. We can't wait to see the flow test results, but once you are happy you have maxed out your design, please confirm these by printing some actual models so we can assess your performance with typical jobs, at different speeds. In conclusion, I would love for teams to understand where your design might sit within the market, based on its performance and cost relative to competitors, and how much we might expect to make off a small production run.

Thanks all, really excited for this next phase of development!

Scott

